

Project Initialization and Planning Phase

Date	04 July 2026
TeamID	
Project Title	A Comprehensive Measure of Human Development
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to predict the **Human Development Index (HDI)** category of a country using Machine Learning, improving the speed and accuracy of development assessment. The system analyzes important socio-economic indicators to classify countries into **Very High, High, Medium, or Low Human Development**. This enables governments, researchers, policymakers, and development organizations to make informed decisions based on data-driven insights.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning-based Human Development Index Prediction System that accurately predicts a country's HDI category using indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita.
Scope	The project focuses on analyzing socio-economic data and predicting the Human Development Index category using Machine Learning algorithms. It provides an interactive web application that assists governments, researchers, educational institutions, and policymakers in evaluating and comparing human development levels across countries.
Problem Statement	
Description	Assessing the Human Development Index manually requires analyzing multiple socio-economic indicators, making the process time-consuming and complex. Traditional methods do not provide instant predictions or interactive analysis, making it difficult to identify development gaps efficiently.
Impact	An accurate HDI prediction system helps policymakers and development organizations identify areas requiring improvement, supports effective planning, enhances resource allocation, and contributes to better decision-making for sustainable human development.
Proposed Solution	

Approach	The proposed solution uses Machine Learning algorithms trained on historical Human Development Index data to predict a country's HDI category based on Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita.
Key Features	- Machine Learning-based Human Development Index prediction. - Predicts Very High, High, Medium, or Low HDI categories. - Interactive and user-friendly web application. - Fast and accurate prediction using socio-economic indicators. - Supports data-driven decision-making for governments and researchers. - Easy-to-use interface with real-time prediction results.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor or equivalent
Memory	RAM specifications	Minimum 8 GB
Storage	Disk space for data ,models, and logs	Minimum 256 TB SSD
Software		
Frameworks	Python frameworks	Flask/ streamlit
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source, size, format	Human Development Index Dataset (CSV), containing Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita, and HDI Category